Corpora: theories, methods, and applications

*Lexicology and Lexicography — **Course Website***

Dr. Quirin Würschinger, LMU Munich

November 25, 2025

Outline

1. **Introduction to corpus linguistics**: Core concepts, methods, and applications of corpus-based research
2. **Sketch Engine tutorial**: Hands-on introduction to key corpus analysis tools
3. **Practice using Sketch Engine**: Group work analysing lexical patterns in BNC 2014 Spoken

Fundamentals of corpus linguistics

What is corpus linguistics about?

Corpus linguistics is a **research methodology** that focuses on the **systematic study of language** using large and diverse **collections of authentic texts**, known as **corpora**.

These collections of language data, either written or spoken, provide an **empirical basis** for analysing:

- **language use** (e.g. collocational patterns such as *pretty woman*)
- **linguistic variation** across different text types or communities (e.g. neologisms such as *smash*)
- **language change** (e.g. *going to* future)

What corpora have you heard of or used? (e.g. COCA, BNC, Google Books)

Goals and approach

For a clear and accessible book-length introduction to corpus linguistics, see Stefanowitsch (2020).

- The primary goal of corpus linguistics is to investigate **linguistic phenomena and patterns** by examining **real-world language usage**.
- This approach contrasts with more traditional linguistic methods that rely heavily on **introspection** and **theoretical speculation** (e.g. Chomsky).
- Corpus linguistics has gained significant momentum in recent years, thanks to advances in **data** (e.g. social-media and web corpora) and **methods** (e.g. social-network analysis, machine learning).
- From a *usage-based* perspective, **linguistic structure and knowledge emerge from the patterns** that speakers encounter in their experience with language.

What is corpus linguistics good for?

What advantages might corpus linguistics offer over introspective methods? Consider:

- What kinds of questions can we answer with corpus data that we couldn't answer (well) otherwise?
- What are potential limitations or challenges?

Advantages of corpus linguistics

- **Authentic language data**: grounded in real-world language use, not idealised examples
- **Quantitative analysis**: frequency counts and statistical measures reveal patterns
  → e.g. *going to* vs *will* for future tense across text types
- **Collocations and constructions**: identify form-meaning pairings in large datasets
  → e.g. *strong tea* vs *powerful tea*; *make a decision* vs *take a decision*
- **Variability and context sensitivity**: examine language across genres, registers, and contexts
  → e.g. *purchase* vs *buy* in formal vs informal contexts
- **Language change**: track diachronic and synchronic patterns
  → e.g. rise of *selfie* over time; decline of *telephone* in favour of *phone*
- **Data-driven applications**: inform language teaching and learning materials
  → e.g. vocabulary lists based on corpus frequency; collocation dictionaries

Key concepts

Corpus types

A **corpus** is a large, **structured collection of texts** that serves as the basis for **linguistic analysis**.

What kinds of research questions could you answer with a corpus that includes many text types versus one focused on a specific domain?

Examples

- **General**: BNC, COCA, Google Books — wide variety of text types
  → e.g. How does the frequency of *going to* vs *will* vary across different text types?
- **Specialised**: BNC 2014 Spoken, academic journal corpora, learner corpora — specific genres or domains
  → e.g. What collocations do learners use with *make* vs *do* in academic writing?

Annotation refers to the process of adding **metadata** or **linguistic information** to a corpus, such as **part-of-speech tags**, **syntactic structure**, or **semantic roles**.

Metadata: data **about** language use on several levels:

- corpus
  - → texts: author, text type, register, topic
    - ↬ paragraphs or utterances
      - ↬ running words (tokens): word class, lemmatisation
        - ↬ tokens: *walk*, *walks*, *walked*
        - ↬ type, lexeme, lemma: WALK ^{v}

Core tools: concordance, collocations, frequency

- **Concordance**: search for words, phrases, or patterns and display results **in context**
  → e.g. examining all instances of *break down* to distinguish phrasal verb vs compound noun usage
- **Collocation**: **co-occurrence of words** within a specific context or proximity
  → e.g. identifying that *strong* collocates with *tea* but *powerful* does not
- **Frequency analysis**: counting occurrences of linguistic features to identify patterns and trends
  → e.g. comparing frequency of *selfie* vs *self-portrait* across different time periods

What might we discover by comparing the collocations of *pretty* vs *beautiful*?

Sketch Engine tutorial

https://wuqui.github.io/SkEtut/

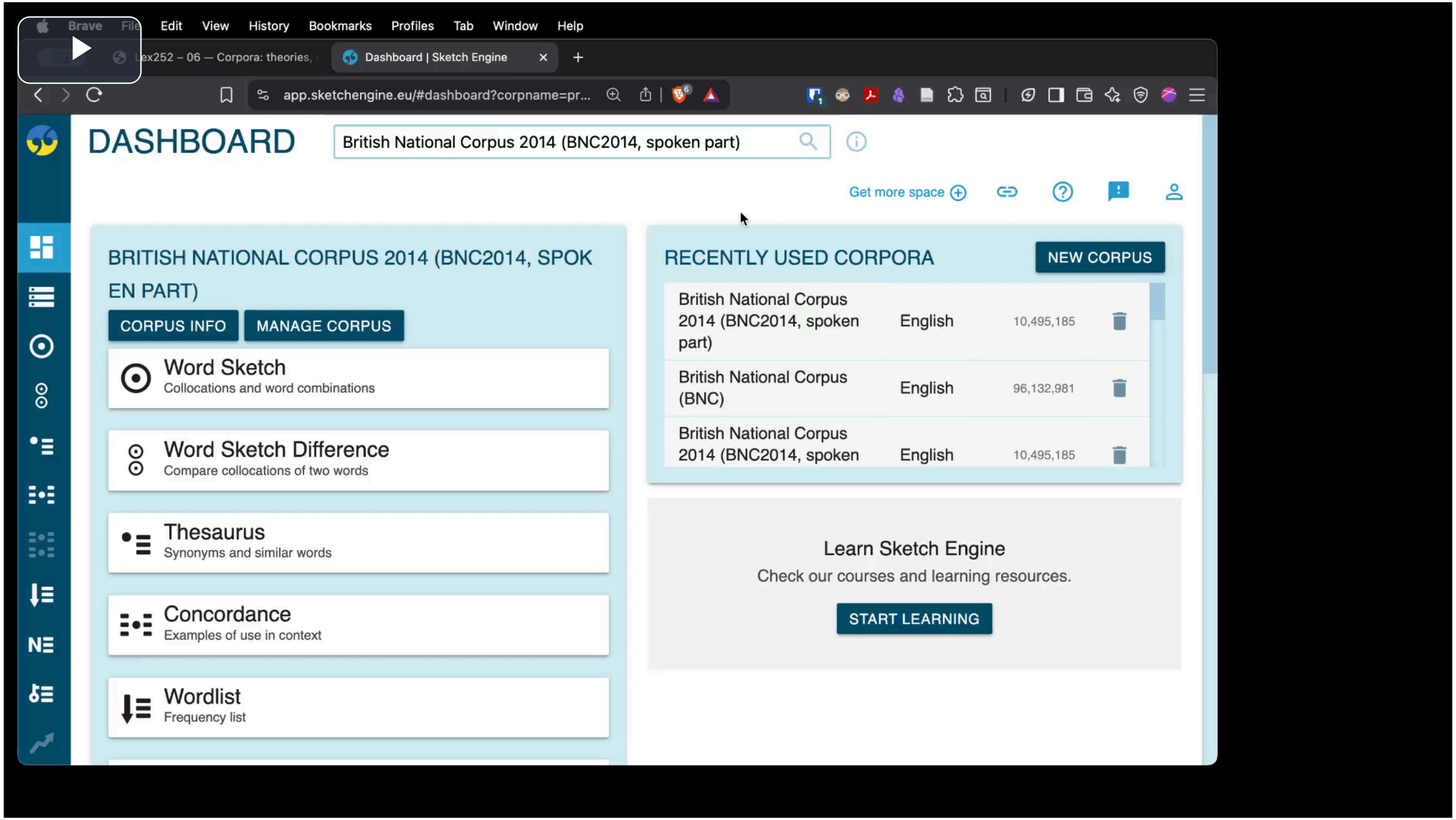
Practice: using Sketch Engine for lexicology

Example analysis in BNC 2014 Spoken

Steps we will walk through together:

- Open **BNC 2014 Spoken** and explore the **corpus information** (size, time period, kinds of spoken data, available **text types** / metadata such as age, gender, education).
- Run **simple concordances** for two items (e.g. *yeah* vs *yes*) and look at a few lines for each to see how they are used.
- Check **overall frequency** for both items and then use **frequency by text type** to see whether their usage differs across text type groups.
- If time allows, briefly look at a **word sketch / collocation view** for each item to see typical lexical partners.

Demo screencast



Task overview

Work in groups with **BNC 2014 Spoken**. There will be **three groups**, each working on a different set of lexical items but following the **same steps**. You have **about 20 minutes** for the corpus work and then **around 3 minutes per group** to report your findings.

During the work phase, each group should:

- explore basic **corpus information** for BNC 2014 Spoken, including which **text types** / metadata categories are available (e.g. age, gender, education)
- run at least one **simple concordance query**
- inspect **basic frequency** (and, if time, simple collocation/word sketch information)
- add notes directly to the shared PowerPoint file that you will use for your report

Task 1: Intensifiers — *really* vs *very*

Question: How are *really* and *very* used as intensifiers in spoken English?

Steps:

1. In Sketch Engine, select the corpus **BNC 2014 Spoken**.
2. Open the **corpus information** view and note down: approximate size, time period, what kinds of spoken data it contains, and which **text types** / metadata dimensions are available (e.g. age, gender, education).
3. Run a **simple concordance** search for lemma *really* and then for lemma *very* (one at a time) in BNC 2014 Spoken. Skim a few lines to see how they are used.
4. Use a **frequency or word list view** (or the frequency information in the concordance results) to compare how frequent *really* and *very* are overall.
5. Choose **one “text type” dimension** (as defined in Sketch Engine), such as age group, education level, or gender, where you **expect differences** between *really* and *very*. Use the **frequency by text type** view in Sketch Engine to compare how frequent each item is across those groups.
6. If time allows, have a quick look at a **word sketch / collocation view** for each item and see what typical collocates appear.

Task 2: Reporting verbs — *say* vs *tell*

Question: How are *say* and *tell* used in spoken English?

Steps:

1. In Sketch Engine, select **BNC 2014 Spoken** (same corpus as Task 1).
2. Look at the **corpus information** and briefly note size, period, main types of spoken interaction, and which **text types** / metadata dimensions are available (e.g. age, gender, education).
3. Run a **simple concordance** search for lemma *say* and then for lemma *tell* (one at a time). Skim several lines for each.
4. Use a **frequency / word list** or concordance frequency information to compare how frequent *say* and *tell* are in BNC 2014 Spoken overall.
5. Choose **one "text type" dimension** (e.g. age, education, gender) where you **expect differences** in how *say* and *tell* are used. Use the **frequency by text type** view in Sketch Engine to compare how frequent each verb is across those groups.
6. If time allows, inspect a **basic collocation / word sketch** view for *say* and *tell* to see what they typically co-occur with (e.g. pronouns, that-clauses).

Task 3: Discourse markers — *like* vs *you know*

Question: How are *like* and *you know* used as discourse markers in spoken English?

Steps:

1. In Sketch Engine, select **BNC 2014 Spoken**.
2. Use the **corpus information** to remind yourselves what kinds of spoken data are included (e.g. everyday conversations, interviews), and which **text types** / metadata dimensions you can use later (e.g. age, gender, education).
3. Run a **simple concordance** search for *like* and then for *you know* (you may have to filter out some non-discourse uses of *like* by reading the lines carefully).
4. Use a **frequency / word list** or frequency information from the concordance to compare how frequent *like* and *you know* are overall.
5. Choose **one "text type" dimension** (e.g. age, education, gender) where you **expect differences** in how *like* and *you know* are used. Use the **text type / metadata breakdown** in Sketch Engine to compare how frequent each item is across those groups.
6. If time allows, look at a **collocation / word sketch** view to see typical patterns (e.g. what comes before/after these items).

References

Stefanowitsch, Anatol. 2020. *Corpus Linguistics*. Berlin: Language Science Press. https://doi.org/10.5281/zenodo.3735822.